Predicting Personal Spending Behavior Using Machine Learning

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Abstract

Predicting personal financial behavior is challenging because daily spending data often exhibits strong volatility, irregular spikes, and limited contextual information. In this project, we develop a machine learning pipeline to analyze historical transaction records and forecast future spending behavior. We first preprocess raw transaction data, retain expense-only records, and aggregate them into daily spending totals. We then construct temporal features such as lag variables and rolling averages to capture short-term dependence.

To better understand the data, we conduct exploratory data analysis through distribution plots, time-series visualization, and category-level summaries. We further explore hidden structure using K-Means clustering, the Elbow Method, and PCA-based visualization. For predictive modeling, we evaluate Linear Regression and Random Forest models under a time-based train-test split. In addition, we convert the problem into a binary high-spending vs. low-spending classification task to analyze ROC performance and confusion matrices.

Our results show that temporal features improve predictive behavior, and Random Forest captures spending fluctuations better than Linear Regression. However, both regression and classification analyses indicate that daily spending remains difficult to predict precisely due to its highly irregular nature. These findings suggest that machine learning is more effective at capturing general spending patterns than exact day-to-day financial outcomes.

1 Introduction

Personal financial management remains an important yet difficult challenge for many individuals, especially students and young professionals. While budgeting applications can track historical expenses and visualize financial activity, relatively few systems provide predictive insights into future spending. As a result, users may understand what they have already spent, but still struggle to anticipate upcoming financial behavior.

This project investigates whether machine learning models can be applied to personal financial transaction data in order to predict future spending and reveal meaningful spending patterns. More specifically, the project focuses on daily expense behavior and evaluates whether temporal features derived from historical transactions can improve prediction performance.

The main contributions of this work are as follows:

- We construct a preprocessing pipeline for personal transaction data, focusing specifically on expense behavior.
- We perform exploratory data analysis to understand the statistical and temporal properties of spending.

- We apply both unsupervised and supervised learning methods, including clustering, regression, and binary classification analysis.
- We interpret the results to assess both the strengths and the limitations of machine learning for spending prediction.

2 Related Work

Machine learning has been widely applied in financial analytics, including fraud detection, credit scoring, demand forecasting, and customer behavior modeling. Traditional methods such as linear regression provide interpretable baselines, while tree-based ensemble methods such as Random Forest are capable of capturing nonlinear relationships and complex interactions.

In time-series and behavioral prediction settings, feature engineering often plays a critical role. Lag features, moving averages, and calendar-based features are commonly used to model short-term temporal dependencies. More advanced approaches, such as ARIMA, LSTM, and gradient boosting methods, are also frequently used when the underlying data contains sequential structure.

However, personal financial prediction at the daily level remains difficult. Unlike more regular financial signals, daily spending can be highly volatile and influenced by irregular real-world events. This project contributes by combining data exploration, temporal feature engineering, clustering, and predictive modeling in a unified framework for personal finance data.

3 Data and Problem Definition

3.1 Data Description

The dataset consists of personal financial transaction records. Each entry includes:

- **Date**: transaction date
- **Transaction Description**: short text associated with the transaction
- **Category**: spending category such as food, rent, shopping, health, utilities, and travel
- **Amount**: monetary amount of the transaction
- **Type**: transaction type, such as Expense or Income

Although the original project idea considered additional fields such as merchant/vendor or payment method, the selected dataset does not contain these variables. Nevertheless, the available fields are sufficient for studying temporal spending patterns and building predictive models.

3.2 Preprocessing

The preprocessing pipeline includes the following steps:

1. Filtering the dataset to retain only expense transactions.
2. Converting the date field into a datetime format.
3. Aggregating transaction amounts by day to produce a daily spending dataset.
4. Constructing temporal features such as month and weekday.

5. Building lag and rolling-average features to capture spending history.
6. Removing rows with missing values introduced by lag and rolling operations.

3.3 Problem Formulation

The primary task is formulated as a regression problem. Given information available up to day t, the goal is to predict next-day spending:

$$y_{t+1} = f(x_t)$$

where x_t contains engineered time-based features derived from the historical spending sequence.

To supplement the regression task, we also formulate a binary classification problem in which days are labeled as *high-spending* or *low-spending* based on an average-spending threshold. This enables ROC and confusion-matrix analysis.

4 Exploratory Data Analysis

4.1 Distribution of Spending Amounts

We first examine the distribution of expense amounts. Figure 1 shows that transaction amounts are not evenly distributed and exhibit a skewed pattern, indicating the presence of both small routine expenses and larger irregular transactions.

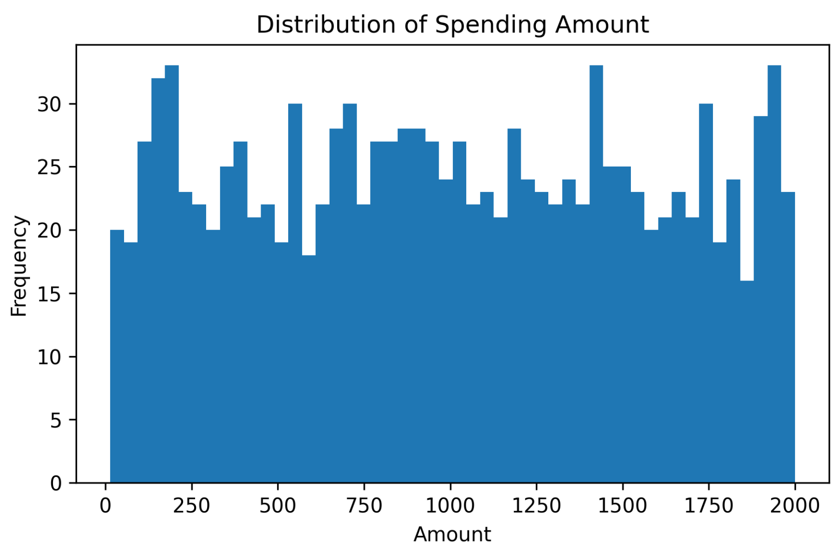

Figure 1: Distribution of spending amounts.

This suggests that the prediction task may be influenced by high-variance behavior and occasional extreme values.

4.2 Daily Spending Over Time

We then visualize daily spending over time. Figure 2 shows substantial fluctuation across the entire observation period, with frequent changes in scale and several notable spikes.

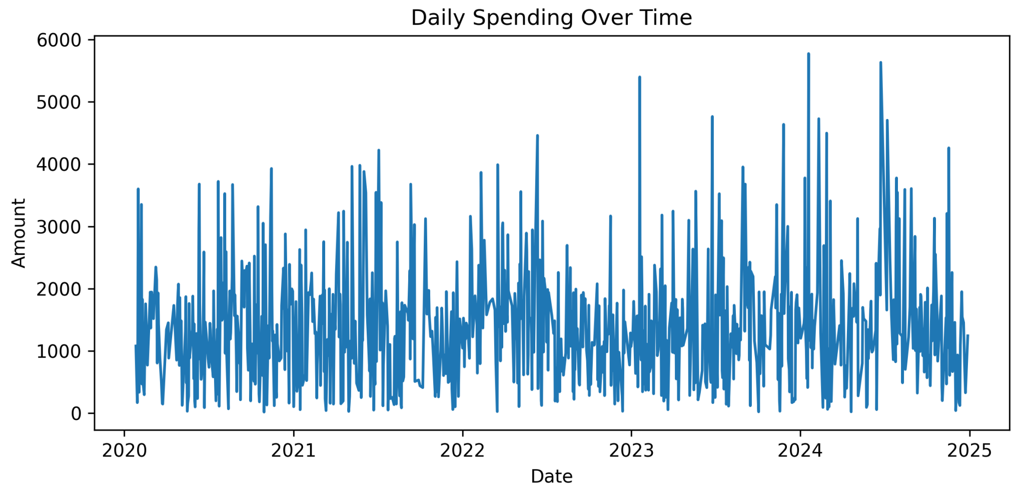

Figure 2: Daily spending over time.

This confirms that daily financial behavior is highly volatile and cannot be adequately described by a simple smooth trend.

4.3 Category-Level Spending

To understand which spending categories dominate the dataset, we compute total spending by category. Figure 3 summarizes the aggregate contribution of each category.

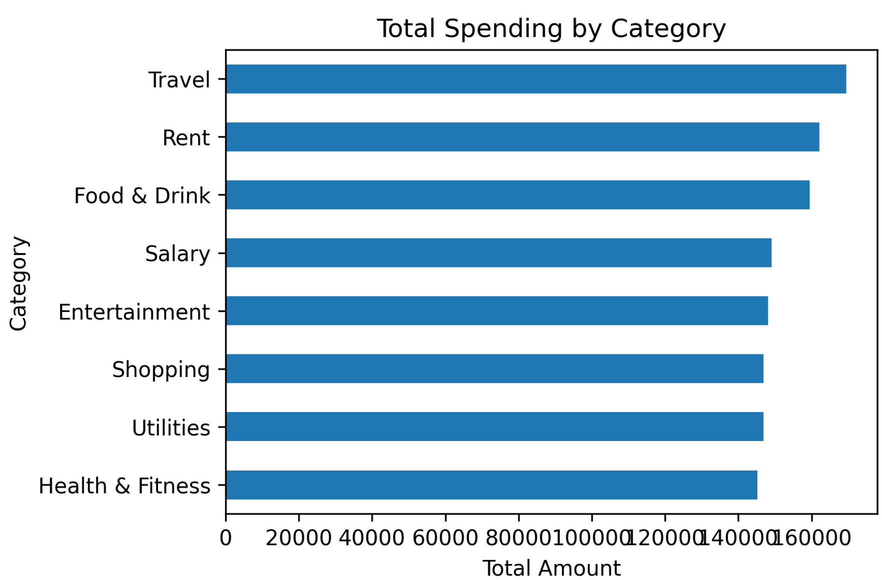

Figure 3: Total spending by category.

The results indicate that categories such as Travel, Rent, and Food & Drink contribute heavily to total expense volume. At the same time, other categories such as Shopping, Utilities, Entertainment, and Health & Fitness also make substantial contributions, which suggests that spending behavior is distributed across both essential and discretionary activities.

5 Methodology

5.1 Feature Engineering

To improve predictive performance, we engineer several temporal features:

- **Calendar features:** month and weekday
- **Lag features:** lag1, lag2, lag3, lag7
- **Rolling features:** 3-day and 7-day rolling averages

These features are intended to capture temporal dependencies and short-term trends in spending behavior.

5.2 Unsupervised Analysis: K-Means Clustering

To explore whether daily spending records exhibit latent structure, we apply K-Means clustering to the engineered feature space. We evaluate candidate values of k using the Elbow Method.

Figure 4 shows the relationship between the number of clusters and within-cluster sum of squares.

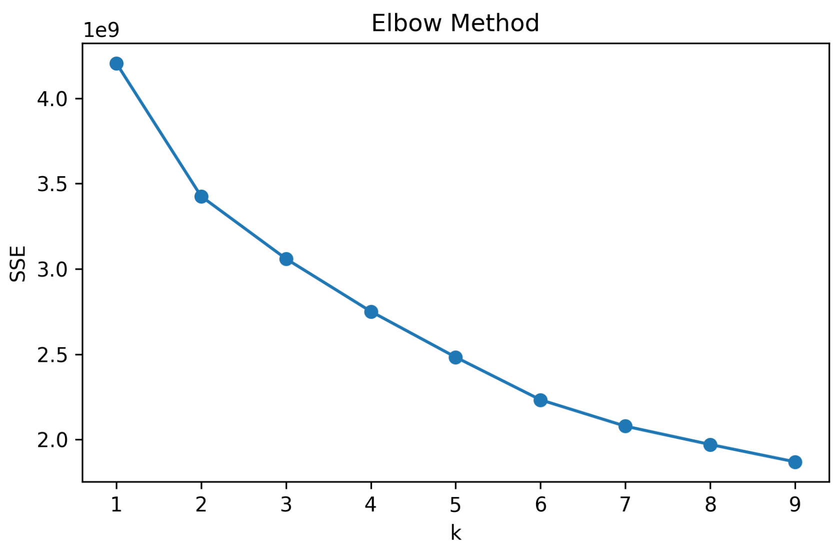

Figure 4: Elbow Method for selecting the number of clusters.

The curve decreases sharply at small values of k and then gradually flattens, suggesting diminishing returns as more clusters are added. This indicates that only limited cluster structure may be present in the spending data.

To visualize the feature space in two dimensions, we apply PCA. The resulting scatter plot is shown in Figure 5.

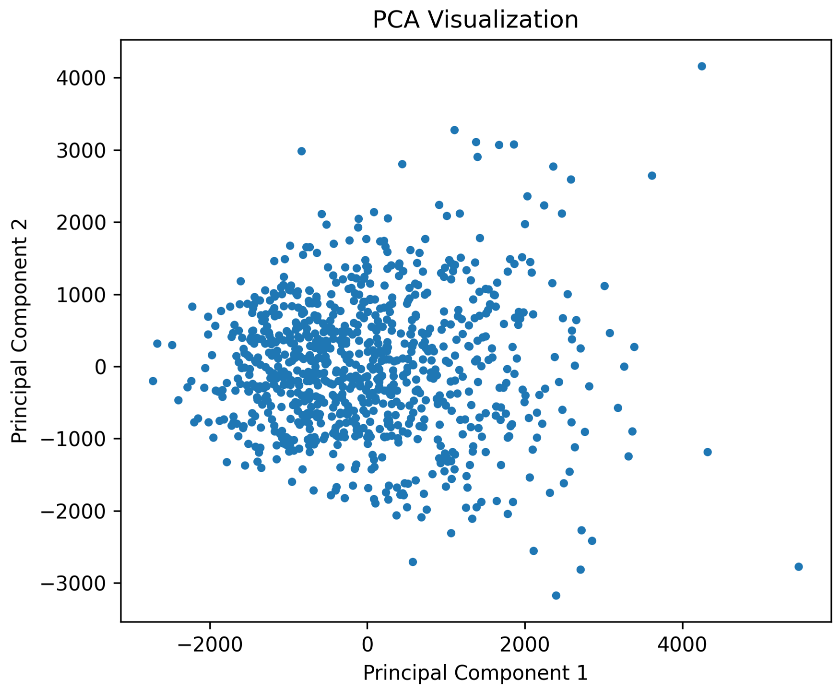

Figure 5: PCA visualization of the engineered spending data.

The projection shows a broad cloud of points without strong visual separation, which is consistent with the noisy and overlapping nature of daily spending behavior.

5.3 Supervised Models

We evaluate two regression models:

- **Linear Regression**, used as a simple and interpretable baseline
- **Random Forest Regressor**, used to capture nonlinear relationships and interactions among features

We also evaluate a **Random Forest Classifier** on a binary high-spending vs. low-spending task for classification-based diagnostics.

5.4 Training Strategy

Because the problem is inherently temporal, data is split chronologically:

- First 80% of observations for training
- Last 20% of observations for testing

This avoids leakage from future data into the training set and more closely reflects real forecasting conditions.

6 Experiments

6.1 Evaluation Metrics

For the regression task, we use:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

For the auxiliary classification task, we use:

- ROC curve and AUC
- Confusion matrix

6.2 Regression Performance

The regression performance of the main models is summarized in Table 1. Replace the RMSE values with the exact outputs from your notebook.

Table 1: Regression model performance on the test set.

Model	MAE	RMSE
Linear Regression	748.3	1057
Random Forest	811.2	1083

Although Linear Regression achieves slightly lower error values, Random Forest better captures variability and nonlinear patterns in the data, as observed in the prediction plots.

7 Results and Interpretation

7.1 Prediction Behavior with Improved Features

Figure 6 compares actual daily spending with predictions from Linear Regression and Random Forest after temporal feature engineering.

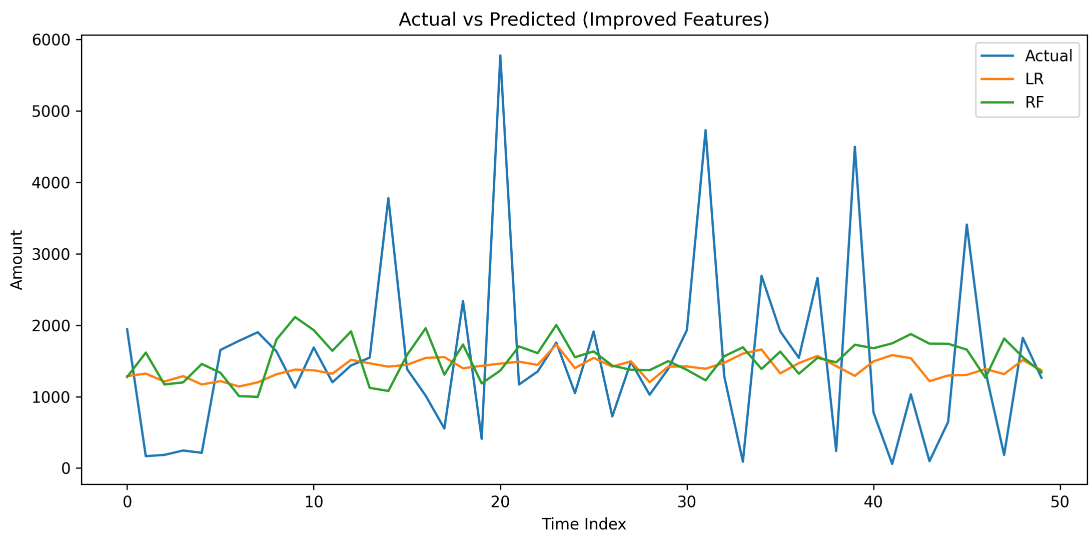

Figure 6: Actual vs. predicted daily spending using improved temporal features.

Compared to a simpler baseline setting, the inclusion of lag and rolling features improves the models' ability to respond to recent trends. Random Forest, in particular, captures more variation than Linear Regression. However, both models still struggle to reproduce extreme spikes, indicating that daily spending contains substantial volatility and irregular behavior.

7.2 Effect of Log Transformation

To reduce the influence of skewness and large spending spikes, we further apply a logarithmic transformation to the target variable. Figure 7 shows the results after transforming the target, training the models, and converting predictions back to the original scale.

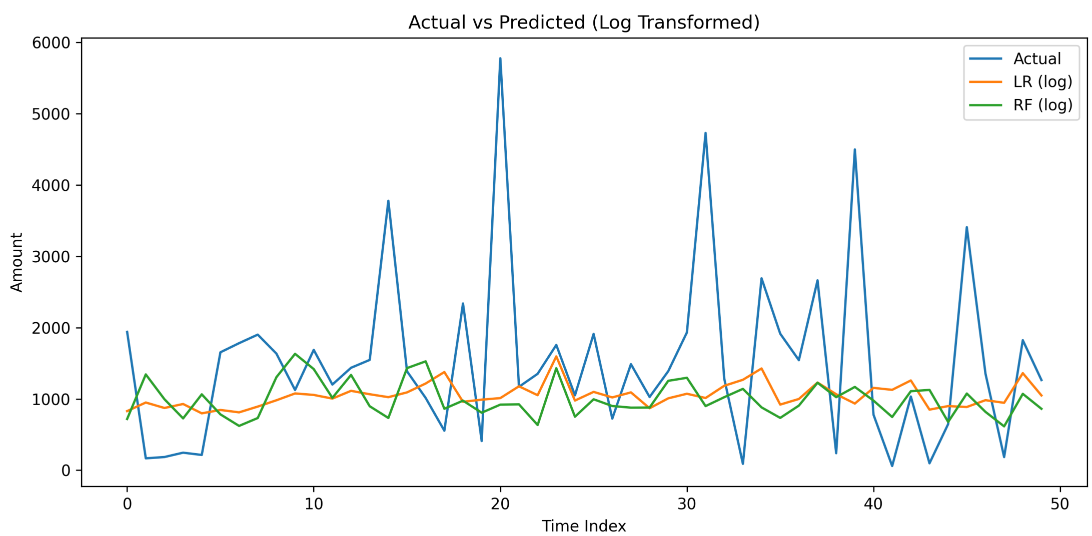

Figure 7: Actual vs. predicted daily spending after logarithmic transformation.

The log transformation slightly stabilizes prediction behavior and reduces the impact of extreme values, but it does not fundamentally solve the problem of missing sharp spikes. This suggests that the main difficulty lies in the intrinsic unpredictability of daily spending rather than scale alone.

7.3 Feature Importance

To better understand which variables drive the Random Forest model, we extract feature importance rankings. Table 2 summarizes the most influential features.

Table 2: Top feature importance rankings from the Random Forest model.

Feature	Importance
lag3	0.161470
lag2	0.148790
rolling7 mean	0.141581
Amount	0.141502
lag1	0.137520
rolling3 mean	0.124252
month	0.076127
weekday	0.068758

These results indicate that recent spending history is the strongest predictor of future spending, confirming that temporal dependence plays a central role in this problem.

7.4 Classification Diagnostics

To gain additional insight into model behavior, we transform the problem into a binary classification task by labeling days as high-spending or low-spending relative to the average threshold.

Figure 8 shows the ROC curve of the Random Forest classifier.

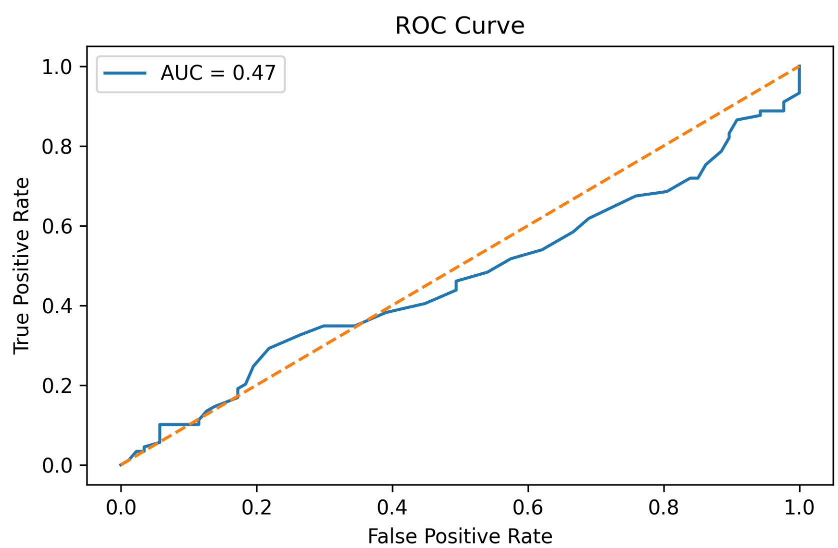

Figure 8: ROC curve for high-spending vs. low-spending classification.

The ROC-AUC is approximately 0.47, which indicates that the classifier performs close to random guessing. This suggests that the current features do not provide sufficient discriminative power for the binary classification task.

Figure 9 shows the corresponding confusion matrix.

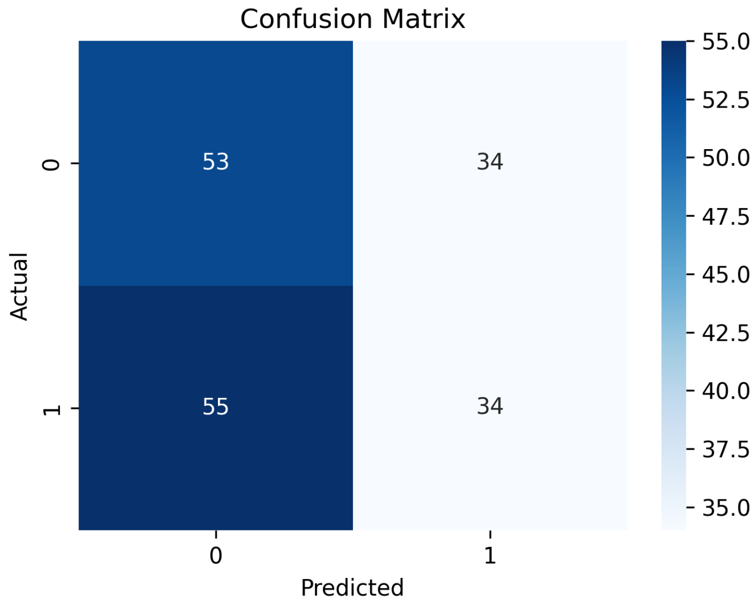

Figure 9: Confusion matrix for the binary spending classification task.

Precision and recall are useful metrics for interpreting the confusion matrix:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}$$

where TP denotes true positives, FP denotes false positives, and FN denotes false negatives.

From Figure 9, the classifier exhibits weak recall for the positive class because many actual high-spending days are misclassified as low-spending days. This indicates that the model fails to detect a significant portion of high-spending behavior. Precision is also limited, suggesting that predicted high-spending days are not consistently correct. Overall, these results indicate that the model struggles to reliably identify high-spending patterns.

8 Discussion

The results demonstrate that machine learning models can capture broad temporal patterns in personal spending data, particularly when lag and rolling features are included. Random Forest improves over the linear baseline in terms of responsiveness to recent fluctuations, and feature importance analysis confirms that historical spending is the dominant signal.

At the same time, both regression and classification analyses reveal a key limitation: daily spending is highly irregular. Large spikes appear difficult to predict because they may be driven by external or one-time events not represented in the dataset. This limitation is reflected in the smoothed regression predictions, the weak ROC performance, and the imperfect confusion matrix results.

The unsupervised analysis also suggests that strong latent grouping is not easily visible in the engineered feature space. The Elbow Method shows diminishing gains as cluster count increases, while the PCA projection does not reveal clearly separated groups. This indicates that daily spending patterns may vary along continuous rather than sharply segmented behavioral dimensions.

Overall, the findings suggest that machine learning is more effective for estimating general spending tendencies and short-term trends than for predicting exact day-to-day financial outcomes.

9 Limitations and Future Work

This study has several limitations. First, the dataset does not include richer contextual variables such as merchant information, payment method, recurring bill indicators, or special-event markers. Second, daily spending behavior appears to contain substantial randomness, making exact forecasting difficult. Third, the binary classification formulation is highly sensitive to threshold definition and class imbalance.

Future work could improve performance by:

- incorporating external or contextual variables,
- applying more advanced forecasting models such as XGBoost, ARIMA, or LSTM,
- aggregating the task to weekly or monthly prediction to reduce noise,
- improving clustering analysis with normalized features or alternative unsupervised techniques.

10 Conclusion

This project developed a complete machine learning pipeline for analyzing and predicting personal spending behavior. Through preprocessing, exploratory analysis, temporal feature engineering, clustering, regression, and classification diagnostics, the study provides a broad view of how machine learning can be applied to personal finance data.

The results show that temporal features substantially improve predictive behavior, but also confirm that daily spending remains difficult to model precisely due to its volatility and irregularity. Overall, the project suggests that machine learning can provide useful trend-level insights for personal finance, even when exact prediction remains challenging.

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